VE401 Final Sindy 2015.8

Catagorical Data

multinomial trial

$\Rightarrow$ multinomial distribution

$$f_{X_1X_2,\cdots X_k}(x_1,\cdots,x_k)=\frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$$

* $E[X_i]=np_i$; $Var X_i=np_i(1-p_i)$;

$Cov(X_i,X_j)=-np_ip_j$

* $\boxed{\sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i}}\sim\chi^2_{k-1}$

criteria $\begin{cases}E_i\geq 1 \text{ for all } i\\ E_i\geq 5 \text{ for } 80\% \text{ of } i\end{cases}$

Pearson's χ^2 Goodness-of-Fit Test

$H_0: p_i=p_{i0}$ for all i, $H_1: p_i\neq p_{i0}$

reject H_0 if

$$\chi^2_{k-1-m}=\boxed{\sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i}}>\chi^2_{\alpha,n-1}$$

m: # of parameters estimated

Testing for Independence

contigency table $r\times c=2\times 2$

	black	white	
cat	n_{11}	n_{12}	$n_{1\cdot}$
dog	n_{21}	n_{22}	$n_{2\cdot}$
	$n_{\cdot 1}$	$n_{\cdot 2}$	n

$E_{ij}=\frac{n_{i\cdot}n_{\cdot j}}{n}$

H_0: the catagorizations are independent

$$\chi^2_{(r-1)(c-1)}=\sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$$

Testing for Homogeneity

	black	white	
cat	p_{11}	p_{12}	$p_{1\cdot}$
dog	p_{21}	p_{22}	$p_{2\cdot}$

$E_{ij}=\frac{n_in_j}{n}$

$H_0: p_{1j}=p_{2j}=\cdots=p_{rj}$, $j=1,\cdots,c$.

Regression

$\alpha/\hat{\ }$ (Y)|(X) regresson/i/p

$Y_i:=Y|x_i=\beta_0+\beta_1x_i+\varepsilon_i$

* $\varepsilon_i=Y_i-\mu_{Y_i}$; $E[\varepsilon_i]=0$

estimators: $B_0=\hat{\beta}_0$, $B_1=\hat{\beta}_1$, $B_0+B_1x=\widehat{\mu_{Y|x}}$

estimates: b_0, b_1

$\Rightarrow$ estimated regression line $\hat{\mu}_{Y|x}=b_0+b_1x$

$\Rightarrow y_i=b_0+b_1x_i+e_i$ residual

$$SSE=\sum_{i=1}^{n}e_i^2=\sum_{i=1}^{n}(y_i-(b_0+b_1x_i))^2$$

sum of square error

Least square estimators (LSE)

$$\begin{cases}B_1=\dfrac{n\sum x_iY_i-(\sum x)(\sum Y)}{n\sum x_i^2-(\sum x_i)^2}\\ B_0=\bar{Y}-B_1\bar{x}\end{cases}$$

Mode assumption

① $Y|x\sim N(\mu_{Y|x},\sigma^2)$

② $x_1\neq x_2\Rightarrow Y|x_1, Y|x_2$ independent

$$\boxed{\frac{B_1-\beta_1}{\sigma/\sqrt{\sum(x_i-\bar{x})^2}}},\ \boxed{\frac{B_0-\beta_0}{\sigma\sqrt{\frac{\sum x_i^2}{n\sum(x_i-\bar{x})^2}}}}\sim N(0,1)$$

proved with identities

① $\sum(x_i-\bar{x})=0$ ② $\sum(x_i-\bar{x})(Y_i-\bar{Y})=\sum(x_i-\bar{x})Y_i$

③ $\sum(x_i-\bar{x})(Y_i-\bar{Y})=\frac{1}{n}(n\sum x_iY_i-(\sum x_i)(\sum Y_i))$

④ $\sum(x_i-\bar{x})^2=\sum(x_i-\bar{x})x_i$

⑤ $\sum(x_i-\bar{x})^2=\frac{1}{n}(n\sum x_i^2-(\sum x_i)^2)$

Notions: $S_{xx}=\sum(x_i-\bar{x})^2$, $S_{yy}=\sum(y_i-\bar{y})^2$, $S_{xy}=\sum(x_i-\bar{x})(y_i-\bar{y})$

$\Rightarrow B_1=\frac{S_{xy}}{S_{xx}}$; $SSE=S_{yy}-B_1S_{xy}$

LSE for variance $S^2=\hat{\sigma}^2=\frac{SSE}{n-2}$

$$\chi^2_{n-2}=\frac{(n-2)S^2}{\sigma^2}=\frac{SSE}{\sigma^2}$$

$$\Rightarrow\boxed{\frac{B_1-\beta_1}{S/\sqrt{S_{xx}}}}=T_{n-2}.$$

$100(1-\alpha)\%$ confidence interval for β_1:

$$B_1\pm t_{\alpha/2,n-2}\frac{S}{\sqrt{S_{xx}}}.$$

Test for Significance of Regression

(significant if $\beta_1\neq 0$)

$H_0:\beta_1=0$ $H_1:\beta_1\neq 0$

reject if $|T_{n-2}|=\left|\frac{B_1}{S/\sqrt{S_{xx}}}\right|>t_{\alpha/2,n-2}$

Inference about slope & intercept

$$\boxed{\frac{B_0-\beta_0}{S\sqrt{\sum x_i^2}/\sqrt{nS_{xx}}}}=T_{n-2}$$

confidence interval for β_0.

$$B_0\pm t_{\alpha/2,n-2}\frac{S\sqrt{\sum x_i^2}}{\sqrt{nS_{xx}}}$$

Estimated mean

$$Z=\boxed{\frac{\hat{\mu}_{Y|x}-\mu_{Y|x}}{\sigma\sqrt{\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}}};\ T_{n-2}=\boxed{\frac{\hat{\mu}_{Y|x}-\mu_{Y|x}}{S\sqrt{\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}}}$$

confidence interval for $\mu_{Y|x}$

$$\hat{\mu}_{Y|x}\pm t_{\alpha/2,n-2}S\sqrt{\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}$$

Predicted value

$\hat{Y}|x=\hat{\mu}_{Y|x}=B_0+B_1x$

$$Z=\boxed{\frac{\hat{Y}|x-Y|x}{\sigma\sqrt{1+\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}}}$$

★ $$T_{n-2}=\boxed{\frac{\hat{Y}|x-Y|x}{S\sqrt{1+\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}}}$$

prediction interval for $Y|x$

$$\hat{Y}|x\pm t_{\alpha/2,n-2}S\sqrt{1+\frac{1}{n}+\frac{(x-\bar{x})^2}{S_{xx}}}$$

Repeated measurements

Y_{ij} - the ith observation of $Y|x_i$

$$SSE_{pe}=\sum_{i=1}^{k}\sum_{j=1}^{n_i}(Y_{ij}-\bar{Y}_i)^2=\sum_{i=1}^{k}\sum_{j=1}^{n_i}Y_{ij}^2-\sum_{i=1}^{k}\frac{1}{n_i}\sum_{j=1}^{n_i}Y_{ij}^2$$

$$SSE_{lf}=SSE-SSE_{pe}$$

$$\chi^2_{n-k}=\boxed{\frac{SSE_{pe}}{\sigma^2}},\ \chi^2_{k-2}=\boxed{\frac{SSE_{lf}}{\sigma^2}}$$

Test for lack of fit

H_0: the model is appropriate

reject H_0 if

$$F_{k-2,n-k}=\frac{SSE_{lf}/(k-2)}{SSE_{pe}/(n-k)}>f_{\alpha,k-2,n-k}$$

Correlation

$\mu_{Y|x}=\mu_Y=\rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)$

$$\rho=\frac{Cov(X,Y)}{\sqrt{Var X\,Var Y}}=\frac{\sigma_{XY}}{\sigma_X\sigma_Y}$$

* X, Y independent $\Rightarrow\rho=0$

* $X=\alpha Y+\beta\Leftrightarrow|\rho|=1$

$\widehat{Var}X=\frac{S_{xx}}{n-1}$, $\widehat{Var}Y=\frac{S_{yy}}{n-1}$, $\widehat{Cov}(X,Y)=\frac{S_{xy}}{n-1}$

$\Rightarrow R=\hat{\rho}=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$ coefficient of determination: R^2

Test for correlation

$H_0:\rho=0$ $H_1:\rho\neq 0$ reject H_0 if

$$\frac{B_1}{S/\sqrt{S_{xx}}}=\boxed{\frac{R\sqrt{n-2}}{\sqrt{1-R^2}}}=T_{n-2}>t_{\alpha/2,n-2}$$

$$\boxed{\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right)}\sim N\left(\frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right),\frac{1}{n-3}\right)$$

$H_0:\rho=\rho_0$ $H_1:\rho\neq\rho_0$. reject H_0 if

$Z>z_{\alpha/2}$,

confidence interval for ρ.

$$\left[\frac{1+R-(1-R)e^{2z_{\alpha/2}/\sqrt{n-3}}}{1+R-(1-R)e^{2z_{\alpha/2}/\sqrt{n-3}}},\frac{1+R-(1-R)e^{-2z_{\alpha/2}/\sqrt{n-3}}}{1+R-(1-R)e^{-2z_{\alpha/2}/\sqrt{n-3}}}\right]$$

$$R^2=\frac{S_{yy}-SSE}{S_{yy}}$$

Multiple Linear Regression (Polynomial)

$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + E_i$

$SSE = \sum e_i^2 = \sum (y_i - (b_0 + b_1 x + \cdots + b_p x_i^p))^2$

$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, X = \begin{pmatrix} 1 & x_1 & \cdots & x_1^p \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & \cdots & x_n^p \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, E = \begin{pmatrix} E_1 \\ \vdots \\ E_n \end{pmatrix}$

$\Rightarrow Y = X\beta + E.$

$\hat{\beta} = b = \begin{pmatrix} b_0 \\ \vdots \\ b_p \end{pmatrix}, e = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} \Rightarrow Y = Xb + e.$

$SSE = \langle Y - Xb, Y - Xb \rangle = (Y - Xb)^T (Y - Xb)$

minimize $SSE \Rightarrow b = (X^T X)^{-1} X^T Y.$

Multilinear

$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + E_i$

$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{p1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & \cdots & x_{pn} \end{pmatrix}$ 其余同上.

Error Analysis

$S_{yy} = \sum (Y_i - \bar{Y})^2 = \sum Y_i^2 - \frac{1}{n}(\sum Y_i)^2$

$= \langle Y, Y \rangle - \langle Y, PY \rangle = \langle Y, (1_n - P) Y \rangle$

quadratic form.

$P = \frac{1}{n}\begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}$. $P^2 = P$, $P^T = P$

↳ orthogonal projection

$(1_n - P)$ is an orthogonal projection

$\operatorname{ran} P := \operatorname{span}\{x_1\}$, $x_1 = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$

$SSE = \langle Y - Xb, Y - Xb \rangle = \langle (1_n - H) Y, (1_n - H) Y \rangle$

$H = X(X^T X)^{-1} X^T$, $\hat{Y} = HY$

hat matrix (o.p.) $HX = X$, $H^T = H$, $H^2 = H$

$(1_n - H) X = 0$ $(1_n - H)^T = 1_n - H$ $(1_n - H)^2 = 1_n - H$.

$\Rightarrow SSE = \langle Y, (1_n - H) Y \rangle$

$\Rightarrow S_{yy} = \underbrace{\langle Y, (1_n - H) Y \rangle}_{=SSE} + \underbrace{\langle Y, (H - P) Y \rangle}_{=:SSR}$

$SSR = \sum (\hat{Y}_i - \bar{Y})^2.$

- SSE: deviation of response from model
- SSR: variation of response predicted
- S_{yy}: total variation

$R^2 = \frac{SSR}{S_{yy}}$

$S_{yy} = SSE + SSR$: $\sum (Y_i - \bar{Y})^2 = \sum (Y_i - \hat{Y})^2 + \sum (\hat{Y}_i - \bar{Y})^2$

---

* $P: \mathbb{R}^2 \to \mathbb{R}^2$. projection. eigenvalue

$Pv = \lambda v \Rightarrow \lambda = 0$ or 1

* $\operatorname{tr} A := \sum_{i=1}^{n} a_{ii}$.

$\operatorname{tr} H = p + 1$. $\Rightarrow \operatorname{tr}(1_n - H) = n - p - 1$

* $\exists$ orthogonal matrix U

(whose columns are eigenvectors of $1_n - H$) such that

$U(1_n - H)U^T = \begin{pmatrix} 1_{n-p-1} & 0 \\ 0 & 0 \end{pmatrix} =: D_{n-p-1}$

$\Rightarrow SSE = \langle E, (1_n - H) E \rangle$

$\frac{SSE}{\sigma^2} = \langle Z, (1_n - H) Z \rangle$

$\chi^2_{n-p-1} = \boxed{\frac{SSE}{\sigma^2}}$

$\hat{\sigma}^2 = S^2 = \frac{SSE}{n-p-1}$

$SSR = \langle Xb, Y \rangle - \frac{1}{n}(\sum_{i=1}^{n} Y_i)^2$

$= \begin{cases} b_0 \sum Y_i + \sum b_j \sum x_{ji} Y_i - \frac{1}{n}(\sum Y_i)^2 & \text{multi} \\ b_0 \sum Y_i + \sum b_j \sum x_i^j Y_i - \frac{1}{n}(\sum Y_i)^2 & \text{poly} \end{cases}$

F-Test for Significance of Regression

$H_0: \beta_1 = \beta_2 = \cdots = \beta_p = 0$, $H_1: \beta_j \neq 0$. reject H_0

if $F_{p,n-p-1} = \boxed{\frac{SSR/p}{SSE/(n-p-1)} = \frac{SSR/p}{S^2}} > f_{\alpha,p,n-p-1}$.

$\boxed{= \frac{n-p-1}{p} \frac{R^2}{1-R^2}}$

$Var\, Y = \begin{pmatrix} Var Y_1 & Cov(Y_2, Y_1) & \cdots & Cov(Y_1, Y_n) \\ Cov(Y_1, Y_2) & Var Y_2 & \ddots & \vdots \\ \vdots & & \ddots & \ddots \\ Cov(Y_1, Y_n) & \cdots & Cov(Y_{n-1}, Y_n) & Var Y_n \end{pmatrix}$

$Var(CY) = C\, Var(Y) C^T$

$\Rightarrow Var\, Y = \sigma^2 1_n$. $Var\, \hat{\beta} = \sigma^2 (X^T X)^{-1}$

$\Rightarrow Var\, B_j = C_{jj} \sigma^2$.

the $(j+1)$th element on the diagonal of $(X^T X)^{-1}$.

$\hat{\beta} = b$ follow a normal d. with mean β, and variance $\sigma^2 (X^T X)^{-1}$

$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{C_{jj}}}$, $T = \frac{\hat{\beta}_j - \beta_j}{S \sqrt{C_{jj}}}$

Confidence interval for β_j:

$\hat{\beta}_j \pm t_{\alpha/2, n-p-1} S \sqrt{C_{jj}}$

Confidence Interval for estimated mean

$\hat{\mu}_{Y|x_0} = x_0^T (X^T X)^{-1} X^T Y.$

$E[\hat{\mu}_{Y|x_0}] = x_0^T \beta$

$Var[\hat{\mu}_{Y|x_0}] = x_0^T (Var\, \hat{\beta}) x_0$

$= \sigma^2 x_0^T (X^T X)^{-1} x_0$

$Z = \boxed{\frac{\hat{\mu}_{Y|x_0} - \mu_{Y|x_0}}{\sigma \sqrt{x_0^T (X^T X)^{-1} x_0}}}$

$T_{n-p-1} = \boxed{\frac{\hat{\mu}_{Y|x_0} - \mu_{Y|x_0}}{S \sqrt{x_0^T (X^T X)^{-1} x_0}}}$

Confidence interval for $\mu_{Y|x}$

$\hat{\mu}_{Y|x} \pm t_{\alpha/2, n-p-1} S \sqrt{x_0^T (X^T X)^{-1} x_0}$

T-test for model sufficiency

$\mu_{Y|x} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$

$H_0: \beta_j = 0$ $H_1: \beta_j \neq 0$. reject H_0 if

$|T_{n-p-1}| = \left| \frac{\hat{\beta}_j}{S \sqrt{C_{jj}}} \right| > t_{\alpha/2, n-p-1}$

Partial F-Test for Model Sufficiency

H_0: the reduced model is sufficient

H_1: the full model is needed.

$F_{p-m, n-p-1} = \frac{n-p-1}{\underset{f}{p} - \underset{r}{m}} \frac{SSE_r - SSE_f}{SSE_f}$

Indicator variable for qualitative predictors

$X = \begin{cases} 1, & \text{cat} \\ 0, & \text{dog} \end{cases}$

Model Selection

Forward Selection Method

一个一个 variable 往上加.

每加一个选 R^2 最大的跟原来的比较

然后 test significance

Backward Elimination Method

一个一个 variable 往下减

每减一个选 R^2 最大的跟原来的比较

然后 test significance

Stepwise Method

Forward Selection 变种.

一个一个往上加. 每加一个选 R^2 最大的跟原来的比较. 不但 test 新加 variable 的 significance 还要 test 之前的.

Maximum R^2 Method

small $PRESS = \sum (y_i - \hat{y}_{i})^2$

small $C_p = \frac{SSE(p)}{MSE(k)} + 2p - n$. ($C_p \approx p$)

small MSE large R^2

Nonlinear Model

exponential $\mu_{Y|x} = \beta_0 e^{\beta_1 x}$, $x > 0$

$\ln y_i = \ln \beta_0 + \beta_1 x_i + \ln e_i \Leftrightarrow y_i^* = \beta_0^* + \beta_1 x_i + e$

power $\mu_{Y|x} = \beta_0 x^{\beta_1}$, $x > 0$.

$\ln y_i = \ln \beta_0 + \beta_1 \ln x_i + \ln e_i$

$\Leftrightarrow y_i^* = \beta_0^* + \beta_1 x_i^* + e_i^*$

reciprocal $\mu_{Y|x} = \beta_0 + \beta_1 / x$. $x > 0$

$y_i = \beta_0 + \beta_1 x_i^* + e_i$. $x_i^* = \frac{1}{x_i}$